

MH1101 Tutorial 5 (Week 6)

Solution

Reference: Sections 3.2, 3.3 (Lecture Notes)

1. Evaluate the following trigonometric integrals:

(i) $\int \sin^6 x \cos^3 x \, dx$

(v) $\int \sin 3x \sin 6x \, dx$

(ii) $\int_0^\pi \cos^4(2t) \, dt$

(vi) $\int_0^{\pi/4} \tan^4 t \, dt$

(iii) $\int \tan x \sec^3 x \, dx$

(vii) $\int \tan^2 x \cos^3 x \, dx$

(iv) $\int \tan^2 x \sec^4 x \, dx$

Solution

(i) The power of cosine is odd. The strategy is:

- save one $\cos x$ factor
- convert everything else to $\sin x$
- substitute $u = \sin x$ (so $du = \cos x \, dx$).

$$\begin{aligned} \int \sin^6 x \cos^3 x \, dx &= \int \sin^6 x \cos^2 x (\cos x) \, dx \\ &= \int \sin^6 x (1 - \sin^2 x) \cos x \, dx \\ &= \int u^6 (1 - u^2) \, du \\ &= \frac{u^7}{7} - \frac{u^9}{9} + C \\ &= \frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + C \end{aligned}$$

- (ii) Both cosine and sine have even power (4 and 0 respectively). The strategy is to make use of the half-angle formula:

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x).$$

Substitute $x = 2t$:

$$\cos^2 2t = \frac{1}{2}(1 + \cos 4t).$$

So

$$\cos^4 2t = \left(\frac{1}{2}(1 + \cos 4t) \right)^2 = \frac{1}{4}(1 + 2 \cos 4t + \cos^2 4t).$$

Applying the half-angle formula again to $\cos^2 4t$, we have

$$\begin{aligned} \cos^4 2t &= \frac{1}{4}(1 + 2 \cos 4t + \frac{1}{2}(1 + \cos 8t)) \\ &= \frac{3}{8} + \frac{1}{2} \cos 4t + \frac{1}{8} \cos 8t \end{aligned}$$

Hence,

$$\begin{aligned} \int_0^\pi \cos^4(2t) dt &= \int_0^\pi \left(\frac{3}{8} + \frac{1}{2} \cos 4t + \frac{1}{8} \cos 8t \right) dt \\ &= \frac{3\pi}{8} + \frac{1}{2} \left[\frac{\sin 4t}{4} \right]_0^\pi + \frac{1}{8} \left[\frac{\sin 8t}{8} \right]_0^\pi \\ &= \frac{3\pi}{8}. \end{aligned}$$

- (iii) The power of $\tan x$ is odd. The strategy is

- Keep one factor of $\tan x \sec x$.
- Express the rest in terms of $\sec x$.
- Substitute $u = \sec x$ ($du = \tan x \sec x dx$).

$$\begin{aligned}
\int \tan x \sec^3 x \, dx &= \int \sec^2 x (\tan x \sec x) \, dx \\
&= \int u^2 \, du \\
&= \frac{u^3}{3} + C \\
&= \frac{\sec^3 x}{3} + C.
\end{aligned}$$

(iv) The power of $\sec x$ is even. The strategy is

- Keep one factor of $\sec^2 x$.
- Express the rest in terms of $\tan x$.
- Substitute $u = \tan x$ ($du = \sec^2 x \, dx$).

$$\begin{aligned}
\int \tan^2 x \sec^4 x \, dx &= \int \tan^2 x \sec^2 x (\sec^2 x) \, dx \\
&= \int \tan^2 x (1 + \tan^2 x) (\sec^2 x) \, dx \\
&= \int u^2 (1 + u^2) \, du \\
&= \frac{u^3}{3} + \frac{u^5}{5} + C \\
&= \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C.
\end{aligned}$$

(v) Use the Product-to-Sum identity:

$$\sin A \sin B = \frac{1}{2}(\cos(A - B) - \cos(A + B)).$$

$$\begin{aligned}
\int \sin 3x \sin 6x \, dx &= \int \frac{1}{2}(\cos(3x - 6x) - \cos(3x + 6x)) \, dx \\
&= \frac{1}{2} \int \cos(-3x) - \cos 9x \, dx \\
&= \frac{1}{2} \int \cos 3x - \cos 9x \, dx \quad (\text{since } \cos(-3x) = \cos 3x) \\
&= \frac{1}{2} \left(\frac{\sin 3x}{3} - \frac{\sin 9x}{9} \right) + C \\
&= \frac{\sin 3x}{6} - \frac{\sin 9x}{18} + C.
\end{aligned}$$

(vi) Standard strategies do not apply directly. We first convert some tangent terms into secants.

$$\begin{aligned}
\int_0^{\pi/4} \tan^4 t \, dt &= \int_0^{\pi/4} \tan^2 t (\sec^2 t - 1) \, dt \\
&= \int_0^{\pi/4} \tan^2 t \sec^2 t \, dt - \int_0^{\pi/4} \tan^2 t \, dt \\
&= \int_0^{\pi/4} \tan^2 t \sec^2 t \, dt - \int_0^{\pi/4} (\sec^2 t - 1) \, dt \\
&= \int_0^{\pi/4} \tan^2 t \sec^2 t \, dt - \int_0^{\pi/4} \sec^2 t \, dt + \int_0^{\pi/4} 1 \, dt
\end{aligned}$$

Now, the standard strategy applies to the first integral, whereas the last two integrals can be evaluated directly.

Let $u = \tan t$. Then $du = \sec^2 t \, dt$. Also, $t = 0 \implies u = 0$, and $t = \pi/4 \implies u = 1$. So

$$\begin{aligned}
\int_0^{\pi/4} \tan^2 t \sec^2 t \, dt &= \int_0^1 u^2 \, du \\
&= \left[\frac{u^3}{3} \right]_0^1 \\
&= \frac{1}{3}.
\end{aligned}$$

On the other hand, since we know that formula $\int \sec^2 x \, dx = \tan x + C$, we have

$$\int_0^{\pi/4} \sec^2 t \, dt = [\tan t]_0^{\pi/4} = 1 - 0 = 1.$$

Also, it is obvious that

$$\int_0^{\pi/4} 1 \, dt = [t]_0^{\pi/4} = \frac{\pi}{4} - 0 = \frac{\pi}{4}.$$

Hence,

$$\int_0^{\pi/4} \tan^4 t \, dt = \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3}.$$

- (vii) Standard strategies do not seem to apply directly. Let's try to express tangent in terms of sine and cosine. Recall the elementary fact that $\tan x = \frac{\sin x}{\cos x}$.

$$\begin{aligned} \int \tan^2 x \cos^3 x \, dx &= \int \frac{\sin^2 x}{\cos^2 x} \cos^3 x \, dx \\ &= \int \sin^2 x \cos x \, dx \end{aligned}$$

From here, the standard strategy can be used. The power of cosine is odd, so the strategy is

- Keep one factor of $\cos x$.
- Express the rest in terms of $\sin x$.
- Substitute $u = \sin x$ ($du = \cos x \, dx$).

$$\begin{aligned}
\int \tan^2 x \cos^3 x \, dx &= \int \sin^2 x (\cos x \, dx) \\
&= \int u^2 \, du \quad (\text{Substitute } u = \sin x) \\
&= \frac{u^3}{3} + C \\
&= \frac{\sin^3 x}{3} + C.
\end{aligned}$$

□

2. Evaluate the integral.

$$(i) \int \frac{1}{t^2 \sqrt{t^2 - 16}} \, dt$$

$$(iv) \int_0^1 \sqrt{x - x^2} \, dx$$

$$(ii) \int_0^1 \frac{1}{(x^2 + 1)^2} \, dx$$

Hint: Express $\sqrt{x - x^2}$ as $\sqrt{a^2 - (x - b)^2}$.

$$(iii) \int \sqrt{5 + 4x - x^2} \, dx$$

$$(v) \int \frac{1}{\sqrt{x^2 + 2x + 5}} \, dx$$

Hint: Express $\sqrt{5 + 4x - x^2}$ as $\sqrt{a^2 - (x - b)^2}$.

Hint: Express $\sqrt{x^2 + 2x + 5}$ as $\sqrt{(x + b)^2 + a^2}$.

Solution

We shall base our choice of trigonometric substitution on the following table:

Expression	Substitution	Identity involved
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$\sin^2 \theta + \cos^2 \theta = 1$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$\tan^2 \theta + 1 = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\tan^2 \theta + 1 = \sec^2 \theta$

We will not mention the domain for θ explicitly in our solution since it is already given in the table.

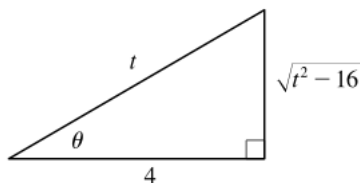
(i) Set $t = 4 \sec \theta$. Then

$$dt = 4 \sec \theta \tan \theta d\theta, \quad \sqrt{t^2 - 16} = \sqrt{16 \sec^2 \theta - 16} = 4 \tan \theta.$$

So

$$\begin{aligned}
 \int \frac{1}{t^2 \sqrt{t^2 - 16}} dt &= \int \frac{1}{16 \sec^2 \theta \cdot 4 \tan \theta} \cdot 4 \sec \theta \tan \theta d\theta \\
 &= \frac{1}{16} \int \frac{1}{\sec \theta} d\theta \\
 &= \frac{1}{16} \int \cos \theta d\theta \\
 &= \frac{1}{16} \sin \theta + C.
 \end{aligned}$$

Next, we need to convert the expression in terms of θ back to that in t . Note that $t = 4 \sec \theta \implies \cos \theta = \frac{4}{t}$, and so we have the following diagram:



Thus,

$$\sin \theta = \frac{\sqrt{t^2 - 16}}{t},$$

Therefore,

$$\int \frac{1}{t^2 \sqrt{t^2 - 16}} dt = \frac{\sqrt{t^2 - 16}}{16t} + C.$$

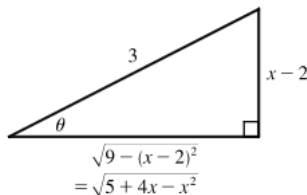
- (ii) Set $x = \tan \theta$. Then $dx = \sec^2 \theta d\theta$, and $(1 + x^2)^2 = (1 + \tan^2 \theta)^2 = \sec^4 \theta$. Also, $x = 0 \implies \theta = 0$, and $x = 1 \implies \theta = \pi/4$. So

$$\begin{aligned} \int_0^1 \frac{1}{(x^2 + 1)^2} dx &= \int_0^{\pi/4} \frac{1}{\sec^4 \theta} \cdot \sec^2 \theta d\theta \\ &= \int_0^{\pi/4} \cos^2 \theta d\theta \\ &= \int_0^{\pi/4} \frac{1}{2} (1 + \cos 2\theta) d\theta \quad (\text{by half-angle formula}) \\ &= \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/4} \\ &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right) \\ &= \frac{\pi}{8} + \frac{1}{4}. \end{aligned}$$

- (iii) We can write (by completing squares):

$$5 + 4x - x^2 = -(x^2 - 4x - 5) = -(x^2 - 4x + 4 - 9) = 3^2 - (x - 2)^2.$$

Let $x - 2 = 3 \sin \theta$. Then $dx = 3 \cos \theta d\theta$.



$$\begin{aligned}
\int \sqrt{5+4x-x^2} \, dx &= \int \sqrt{9-9\sin^2\theta} \cdot 3\cos\theta \, d\theta \\
&= 9 \int \cos^2\theta \, d\theta \\
&= 9 \int \frac{1}{2}(1+\cos 2\theta) \, d\theta \quad (\text{by half-angle formula}) \\
&= \frac{9}{2}\theta + \frac{9}{4}\sin 2\theta + C \\
&= \frac{9}{2}\theta + \frac{9}{2}\sin\theta\cos\theta + C \\
&= \frac{9}{2}\sin^{-1}\left(\frac{x-2}{3}\right) + \frac{9}{2} \cdot \frac{x-2}{3} \cdot \frac{\sqrt{5+4x-x^2}}{3} + C \\
&= \frac{9}{2}\sin^{-1}\left(\frac{x-2}{3}\right) + \frac{x-2}{2} \cdot \sqrt{5+4x-x^2} + C
\end{aligned}$$

(iv) We can write (by completing squares)

$$\sqrt{x-x^2} = \sqrt{\frac{1}{4} - (x^2 - x + \frac{1}{4})} = \sqrt{\left(\frac{1}{2}\right)^2 - (x - \frac{1}{2})^2}.$$

Let $x - \frac{1}{2} = \frac{1}{2}\sin\theta$. Then $dx = \frac{1}{2}\cos\theta \, d\theta$, and

$$\sqrt{x-x^2} = \sqrt{\frac{1}{4} - \frac{1}{4}\sin^2\theta} = \frac{1}{2}\cos\theta.$$

Also, $x = 0 \implies \sin\theta = -1 \implies \theta = -\frac{\pi}{2}$ (since $-\pi/2 \leq \theta \leq \pi/2$). And

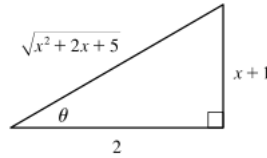
$x = 1 \implies \sin \theta = 1 \implies \theta = \frac{\pi}{2}$ (since $-\pi/2 \leq \theta \leq \pi/2$). So

$$\begin{aligned}
 \int_0^1 \sqrt{x-x^2} \, dx &= \int_{-\pi/2}^{\pi/2} \frac{1}{2} \cos \theta \frac{1}{2} \cos \theta \, d\theta \\
 &= \frac{1}{4} \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta \\
 &= \frac{1}{4} \int_{-\pi/2}^{\pi/2} \frac{1}{2} (1 + \cos 2\theta) \, d\theta \quad (\text{by half-angle formula}) \\
 &= \frac{1}{8} \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2} \\
 &= \frac{1}{8} \left[\left(\frac{\pi}{2} + 0 \right) - \left(-\frac{\pi}{2} + 0 \right) \right] \\
 &= \frac{\pi}{8}.
 \end{aligned}$$

(v) We can write (by completing squares)

$$x^2 + 2x + 5 = (x+1)^2 + 2^2.$$

Set $x+1 = 2 \tan \theta$. Then $dx = 2 \sec^2 \theta \, d\theta$.



$$\begin{aligned}
 \int \frac{1}{\sqrt{x^2 + 2x + 5}} \, dx &= \int \frac{1}{\sqrt{4 \tan^2 \theta + 4}} \cdot 2 \sec^2 \theta \, d\theta \\
 &= \int \frac{2 \sec^2 \theta}{2 \sec \theta} \, d\theta \\
 &= \int \sec \theta \, d\theta \\
 &= \ln |\sec \theta + \tan \theta| + C \\
 &= \ln \left| \frac{\sqrt{x^2 + 2x + 5}}{2} + \frac{x+1}{2} \right| + C.
 \end{aligned}$$

□

3. Evaluate the integral

$$\int \tan^2 x \sec x \, dx.$$

Solution Let $t = \tan x$, $s = \sec x$.

Let $J = \int t^2 s \, dx$. Set $u = t$, $v' = st$. By the integration-by-parts, we have

$$\begin{aligned} J &= ts - \int s^3 \, dx \\ &= ts - \int (1 + t^2)s \, dx \\ &= ts - \int s \, dx - J \\ 2J &= ts - \ln |s + t| \\ J &= \frac{1}{2} (ts - \ln |s + t|) + C \\ &= \frac{1}{2} (\tan x \sec x - \ln |\sec x + \tan x|) + C. \end{aligned}$$

Answer Keys. **1.** (i) $\frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + C$ (ii) $\frac{3\pi}{8}$ (iii) $\frac{\sec^3 x}{3} + C$ (iv) $\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$ (v) $\frac{\sin 3x}{6} - \frac{\sin 9x}{18} + C$ (vi) $\frac{\pi}{4} - \frac{2}{3}$ (vii) $\frac{\sin^3 x}{3} + C$ **2.** (i) $\frac{\sqrt{t^2-16}}{16t} + C$ (ii) $\frac{\pi}{8} + \frac{1}{4}$ (iii) $\frac{9}{2} \sin^{-1} \left(\frac{x-2}{3} \right) + \frac{x-2}{2} \cdot \sqrt{5+4x-x^2} + C$ (iv) $\frac{\pi}{8}$ (v) $\ln \left| \frac{\sqrt{x^2+2x+5}}{2} + \frac{x+1}{2} \right| + C$ **3.** $\frac{1}{2} (\tan x \sec x - \ln |\sec x + \tan x|) + C$ $\square$